Neural Network Components

AIML-501-01B Model Development

Brianna Gliesmann

Network Architecture: How Data Flows

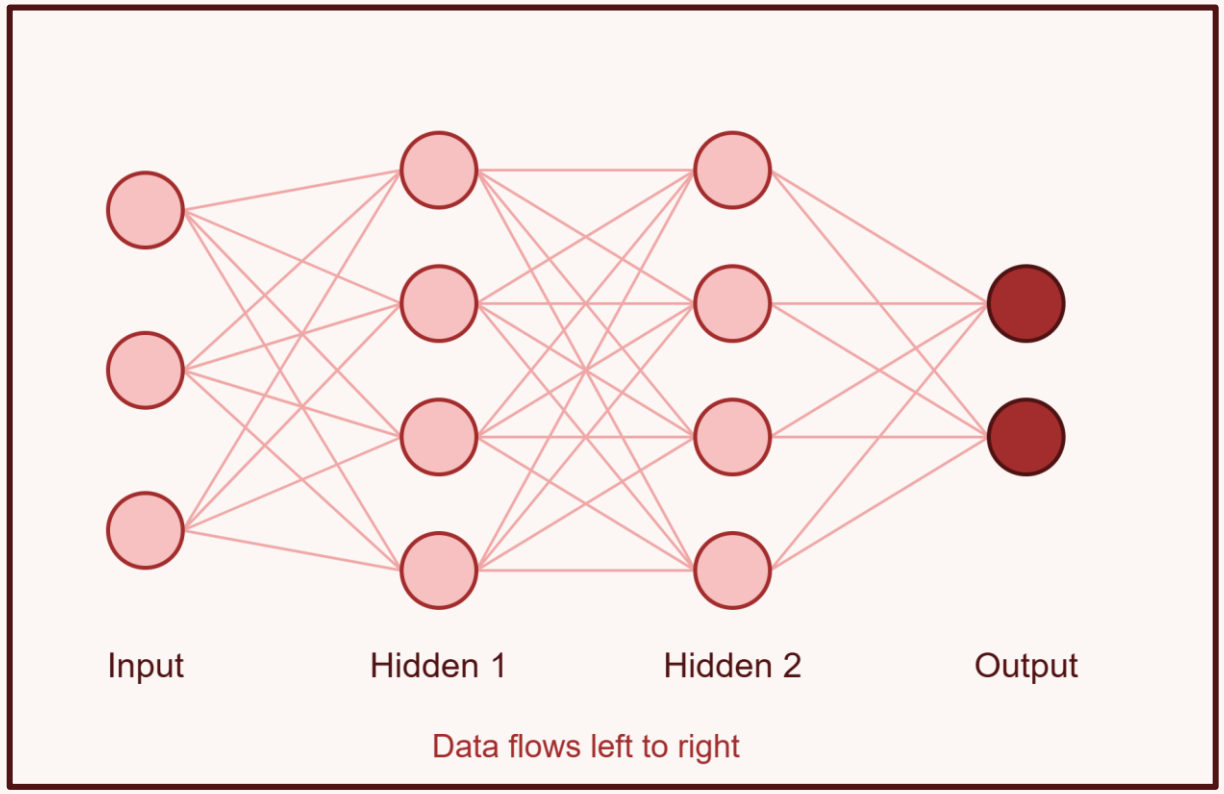


Figure 1: Network Architecture

Layers

Input, hidden and output layers. Each layer's outputs become the next layer's inputs.

Neurons (circles)

Compute a weighted sum plus a bias, then apply an activation function. Dark red neurons=output neurons.

Weights (lines)

Learned numbers that set how strongly one neuron influences the next.

Inside One Neuron

1 **Inputs × weights**

Each input is multiplied by its learned weight. A larger weight means more influence.

→

2 **Sum plus bias**

The weighted inputs are added to a bias, a learned constant that shifts the result.

→

3 **Activation**

A non-linear function (ReLU, sigmoid, tanh) lets the network learn complex patterns.
Adds non-linearity

→

4 **Output**

Passed to every neuron in the next layer or returned as the prediction.

Figure 2: Computation inside one neuron

$$\text{output} = f(w_1x_1 + w_2x_2 + w_3x_3 + b)$$

x = inputs, w = weights, b = bias, f = activation function (shown for three inputs)

Multiply each input by its weight, add the results and the bias, then pass the total through the activation function.

How the Network Learns

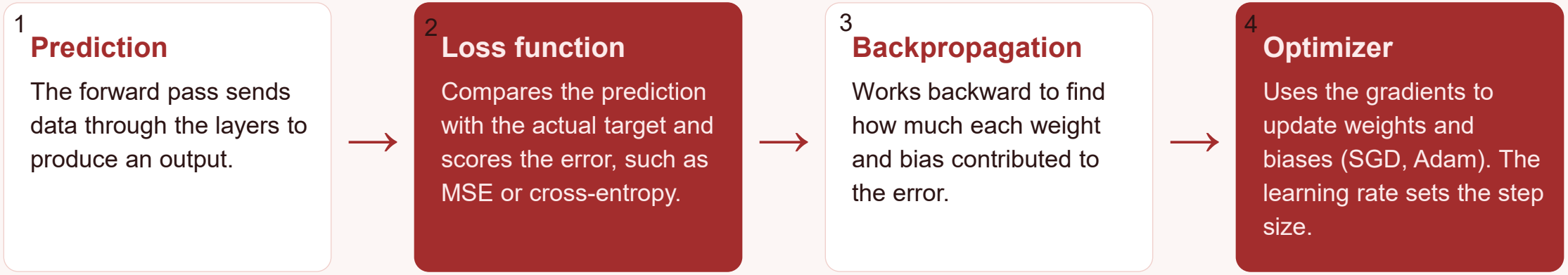


Figure 3: The training loop

Repeat over many examples until the loss stops improving

Classifying the Components

Structure

What the network is made of: data is represented and transformed in the forward pass

- **Layers**: organize neurons into input, hidden and output tiers
- **Neurons**: compute a weighted sum plus a bias
- **Weights**: set connection strength and are learned
- **Activation functions**: add non-linearity

Training

How the network learns: errors are measured and the parameters are updated

- **Loss function**: scores the error against the target
- **Optimization algorithm**: uses gradients from backpropagation to update weights and biases

Rationale: structure components act on the way forward, while training components act on the result. Weights link the two groups, because the optimizer adjusts them.

Reflection on Learning

Challenge

Truly understanding each component, and how they connect into one network.

Strategies

Explaining each component in simpler terms, and writing a clear prompt that described the flowchart I needed.

Knowledge gained

Each piece is simple alone. The network's power comes from how weights, activations, loss and the optimizer work together.

Course connection

Neural Network Playground: adding neurons made training slower, showing how layers and neurons add complexity and training time.

Summary: Why Visualize Neural Networks

Why it matters

- Turns an abstract model into something you can trace from input to output
- Shows how layers, neurons and weights connect
- Separates the forward pass from the training loop

Key insights

- A network is simple units repeated at scale: weighted sum, bias, activation
- Without non-linear activations, stacked layers collapse into one linear model
- Training is a loop: predict, measure loss, backpropagate, update